```
lda_model=create_model('lda')
```

	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
3	0.8182	0.9513	0.8056	0.8182	0.8182	0.7250	0.7250
4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Mean	0.9818	0.9951	0.9806	0.9818	0.9818	0.9725	0.9725
SD	0.0545	0.0146	0.0583	0.0545	0.0545	0.0825	0.0825

Figure 5: Creating the model

```
tuned_lda=tune_model(lda_model)
```

	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
3	0.9091	0.9513	0.9167	0.9318	0.9091	0.8642	0.8750
4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Mean	0.9909	0.9951	0.9917	0.9932	0.9909	0.9864	0.9875
SD	0.0273	0.0146	0.0250	0.0205	0.0273	0.0407	0.0375

Figure 6: Tuning the model

The *compare_models()*, by default, applies ten-fold cross validation and calculates different performance metrics like accuracy, AUC, recall, precision, F1 score, Kappa and MCC for different classifiers with lesser training times, as shown in Figure 4. By passing the tubro=True to *compare_models()* function we can try all the classifiers.

Step 5: Now create the model, as shown in Figure 5:

```
lda_model=create_model ('lda')
```

The Linear Discriminant Analysis classifier is performing well, as shown in Figure 4. So by passing 'lda' to the *create_model()* function, we can fit the model.

Step 6: The next step is to fine tune the model, as shown in Figure 6.

```
tuned_lda=tune_model(lda_model)
```

Tuning of hyper parameters can improve the model accuracy. The *tune_model()* function improved the Linear Discriminant Analysis model accuracy from 0.9818 to 0.9909, as shown in Figure 7.

Step 7: The next step is to make predictions, as shown in Figure 8:

```
predictions=predict_model(tuned_lda)
```

```
print(tuned_lda)

LinearDiscriminantAnalysis(n_components=None, priors=None, shrinkage=0.05,
                           solver='lsqr', store_covariance=False, tol=0.0001)
```

Figure 7: Tuned model details

```
[8] predictions=predict_model(tuned_lda)
```

	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
0	Linear Discriminant Analysis	0.9783	1.0	0.9667	0.9793	0.978	0.9659	0.9667

Figure 8: Predictions using the tuned model